LIMITED-DISCLOSURE RESEARCH CONCEPT

PAPER 1

INVARIANT-SC

Auditable Semantic Recovery for Sustainable and Resilient Supply Chains

CONCEPT IN ONE SENTENCE

Recover what supply-chain records mean before asking AI to optimize them: learn a common, auditable interpretation of heterogeneous enterprise data using physical flow consistency, while preserving evidence and explicitly representing meaning that the observations cannot resolve.

Research identity: one new learning problem, one canonical scientific object, multiple downstream tests. Disclosure level: problem, contribution boundary, evaluation logic, and impact; sensitive mechanics withheld.

1. The Real Problem

Most supply-chain AI begins after records have already been cleaned, joined, and given stable meanings. That assumption is routinely false across organizations. Procurement, warehouse, manufacturing, transport, returns, and finance systems may encode the same physical event with incompatible schemas, units, signs, identifiers, granularity, and posting times. A receipt in one database may be a ledger adjustment in another; one shipment may be split across several tables.

This is not merely an IT inconvenience. If the activity records are interpreted incorrectly, every model built above them inherits the error: Scope 3 activity data becomes difficult to trace, material losses and returns are misclassified, inventory states drift, and disruption paths are reconstructed from an unreliable foundation. NIST similarly identifies cross-domain semantic differences and low-effort traceability-data integration as an unresolved need [5].

HIDDEN PREREQUISITE

Before a system can forecast, optimize, calculate emissions, or plan recovery, it must know what each record means and whether all records can jointly describe a physically plausible process.

Why existing fixes are incomplete

- Manual mappings are expensive, brittle, and difficult to audit when schemas change.
- Language and similarity methods can match plausible labels but cannot by themselves guarantee a globally coherent process [1].
- Standards define useful target representations, yet arbitrary legacy records still need to be reconciled into them [5, 6].
- Relational learning usually assumes the tables, entities, and links are already meaningful [2].
- A dashboard can visualize integrated data; it cannot establish whether the integration is scientifically defensible.

Central research question

PAPER 1 QUESTION

Can an AI system recover a shared supply-chain interpretation from heterogeneous records by exploiting coupled physical-flow regularities, and can it identify when the evidence supports several interpretations rather than one?

2. One Umbrella, One Scientific Object

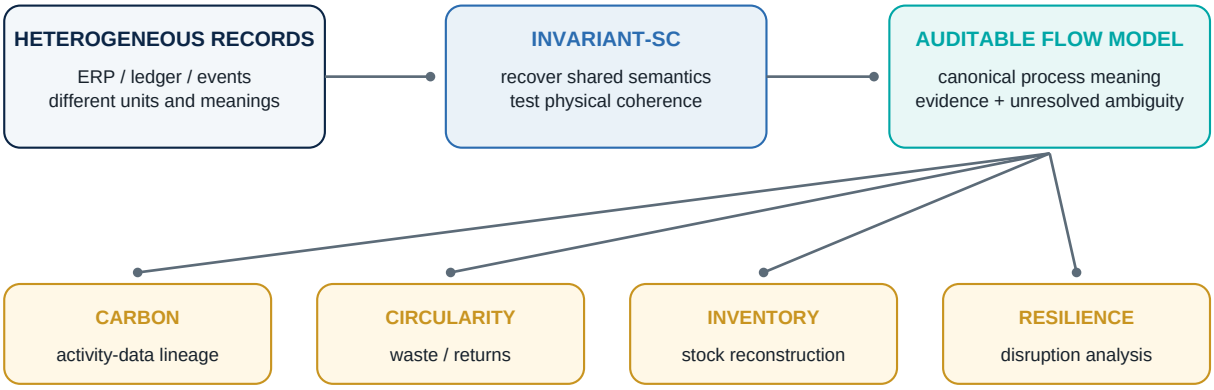

Figure 1. Every application queries the same recovered model; there are no independent feature silos.

2.1 The core

INVARIANT-SC treats data reconciliation as a structured learning problem. It seeks a canonical description of entities, movements, transformations, stocks, timing, and measurement conventions that makes many records jointly coherent. The essential supervision comes from regularities of the physical process, not only from headers or documentation.

2.2 The restrained evidence layer

The recovered model carries source lineage and evidence consistency. When two materially different meanings fit the observations, the system should preserve the alternatives or abstain instead of manufacturing certainty. This evidence-and-ambiguity function strengthens the core claim; it is not a second product or a second headline contribution.

2.3 Why this is a valid umbrella

Downstream question	Query on the shared recovered model	Boundary
Carbon	Which traceable material, energy, and transport activities occurred?	Emission factors and reporting rules stay external.
Waste / circularity	Where did material leave, return, transform, or fail to reconcile?	No claim to solve every circular-economy policy question.
Inventory	What stock trajectory is supported across inconsistent ledgers?	The system exposes unsupported intervals.
Resilience	Which flows and dependencies transmit a disruption?	Decision optimization is a downstream extension.

UNIFYING THESIS

Recover an auditable set of physically consistent canonical supply-chain interpretations from heterogeneous enterprise records, and explicitly separate what is supported, ambiguous, or missing.

3. Where the Novelty Actually Lives

The novelty is not the broad phrase 'AI for supply chains,' and it is not the act of combining carbon and resilience on one screen. It is the proposed learning object and supervision mechanism: infer latent data meaning and process structure from coupled physical consistency, then return an auditable interpretation set rather than a forced one-to-one mapping.

Adjacent direction	What it normally assumes or solves	Paper 1 distinction
Schema matching	Corresponding attributes from names, structure, or examples.	Tests a complete interpretation against coupled flow behavior over time.
Domain-constraint matching	Uses declared schema or domain constraints to improve mappings.	Learns latent process semantics from observed physical-flow consistency and evaluates recovery under shift.
Semantic typing	Assigns known labels to columns, often independently.	Recovers a connected process interpretation, measurement conventions, and relationships.
Relational learning	Learns on databases whose keys and meanings are given [2].	Addresses the upstream recovery of trustworthy common meaning.
Conservation-aware ML	Uses known physical equations to improve prediction [3].	Uses physical regularity to infer the interpretation of the records themselves.
Traceability standards	Specify how interoperable events should be represented [5, 6].	Reconciles heterogeneous legacy evidence toward an auditable canonical view.

A defensible novelty claim

CURRENT POSITION AFTER PRIOR-ART SCREENING

The differentiated contribution is the joint task of physical-invariant-guided semantic recovery, global process consistency, evidence lineage, and explicit non-uniqueness for heterogeneous supply-chain records. The cited fields contain important pieces, but not this complete learning target and evaluation object.

This is a concrete research hypothesis, not an unsupported 'world-first' promise. The novelty becomes publication-grade only if the paper delivers the new task, a method that beats strong alternatives under unseen schemas, a rigorous account of ambiguity, and a reproducible benchmark. Removing any two of these would make the work look like an integration system rather than a top-tier ML contribution.

4. Why This Is Not Another Dashboard

THE KEY DISTINCTION

A dashboard begins after meaning has been supplied. INVARIANT-SC attacks the hidden prerequisite: it learns and audits the representation on which any trustworthy dashboard, optimizer, or carbon analysis would depend.

Dimension	Typical dashboard / platform	Paper 1
Input	Curated, mapped, business-ready data.	Heterogeneous records with uncertain semantics and links.
Core operation	Aggregate, monitor, visualize, alert.	Infer and test shared process meaning.
Output	Metrics, charts, scenarios, recommendations.	Canonical flow interpretation plus lineage and unresolved alternatives.
Failure behavior	Manual exception or silent propagation.	Abstain or expose competing interpretations when evidence is insufficient.
Scientific evaluation	Usability and business KPI improvement.	Recovery, cross-schema generalization, robustness, calibration, and downstream error.
Reusable insight	A deployed application configuration.	A general learning problem, method, theory target, and benchmark.

Proposed paper contributions

1. A benchmarkable task for recovering process semantics under severe schema and organization shift.
2. A learned-and-verified framework that couples local data evidence with global physical-flow consistency.
3. A principled characterization of when observations identify one interpretation and when they do not.
4. A reproducible benchmark with controlled schema variation, missingness, delays, and measurement corruption.
5. Downstream evidence showing how semantic recovery error propagates into carbon, waste, inventory, and resilience analyses.

PRODUCT FEATURES ARE DELIBERATELY EXCLUDED

No procurement chatbot, KPI portal, universal carbon calculator, or multi-agent control room is needed for the first paper. A small visualization may communicate results, but it is not the contribution.

5. Validation Plan and Real-World Value

5.1 What would make the claim credible

The evaluation should test scientific recovery rather than presentation quality. Training and test settings must differ in schemas, naming conventions, process structures, organizations, and data quality. Baselines should include modern language-model matching, semantic typing, join discovery, relational methods, constraint-based alternatives, and simple domain heuristics.

Test	Question answered
Semantic and relational recovery	Did the system reconstruct the intended roles, links, and process meaning?
Unseen-schema generalization	Does physical guidance transfer when labels and database layouts change?
Missing / noisy / delayed records	Does it remain useful without treating every discrepancy as impossible?
Ambiguity and abstention	Does confidence fall when the evidence does not select a unique interpretation?
Downstream error	Do better recovered meanings measurably improve carbon, waste, inventory, and disruption analyses?

5.2 Real issue solved

- Create traceable activity-data lineage before applying Scope 3 calculation methods [7].
- Detect contradictions across procurement, production, warehouse, shipment, return, and waste records.
- Reduce repeated manual remapping when organizations or software systems change.
- Build a shared stock-and-flow basis for shortage, dependency, and disruption studies.
- Show decision-makers exactly where the available evidence is insufficient for a defensible claim.

5.3 Honest limits

- Physical consistency cannot recover information that was never observed; some meanings will remain equivalent.
- Carbon factors, organizational boundaries, allocation rules, and reporting policy remain external to the model.
- The approach does not infer labor conditions, supplier ethics, or every social-sustainability dimension from flow data.
- A synthetic benchmark can establish control and scale, but external datasets or expert validation are still needed for real-world credibility.

VENUE FIT

ICLR fit is strongest if the paper leads with representation learning, structured induction, generalization, and identifiability. If the theory or cross-schema learning claim does not survive, data-management venues may be a more natural home than stretching the work into a dashboard paper.

6. Decision Summary and Selected References

RECOMMENDATION

Proceed with Paper 1 as INVARIANT-SC. Keep one headline contribution: auditable semantic recovery from physical flow consistency. Keep carbon, circularity, inventory, and resilience as coordinated downstream evaluations of the same recovered model. Keep the evidence-and-ambiguity layer subordinate to the core.

What success would look like

1. Reviewers recognize a new ML task, not a repackaged sustainability platform.
2. The system generalizes to unseen data conventions because its supervision is tied to the underlying process.
3. Ambiguous cases are exposed rather than hidden behind confidence scores.
4. A single recovered representation improves several sustainability and resilience analyses without separate bespoke pipelines.

Selected public references

[1] N. Seedat and M. van der Schaar. Bootstrapping Self-Improvement of Language Model Programs for Zero-Shot Schema Matching. ICML, 2025. PMLR page.

[2] M. Fey et al. Position: Relational Deep Learning - Graph Representation Learning on Relational Databases. ICML, 2024. PMLR page.

[3] D. Hansen et al. Learning Physical Models that Can Respect Conservation Laws. ICML, 2023. PMLR page.

[4] S. Chang et al. Learning Production Functions for Supply Chains with Graph Neural Networks. AAAI, 2025. AAAI page.

[5] NIST. Enabling and Using Traceability and Data Linking for Sustainable and Efficient Supply Chains. Project page.

[6] GS1. EPCIS Standard, version 2.0.1. Official standard.

[7] GHG Protocol. Scope 3 Calculation Guidance. Official guidance.

DISCLOSURE NOTE

This faculty brief intentionally omits formulas, the internal program representation, exact learning and verification procedures, proof strategy, benchmark-generation recipes, thresholds, private results, and the implementation roadmap.

Novelty statement current to 2 September 2026: the proposal is strongly differentiated at the level of the joint learning task described here; final novelty and venue claims remain contingent on a systematic review, successful implementation, and comparative evidence.